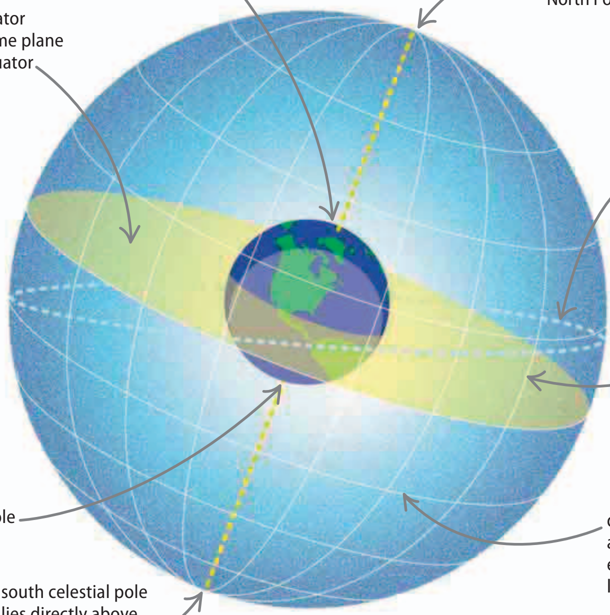

Celestial sphere

The objects seen in the night's sky are not all the same distance from Earth. The Moon is obviously much closer than Jupiter, but astronomers plot their movements and the positions of all the stars on an imaginary sphere that surrounds Earth. The view of the celestial sphere, as it is called, changes as Earth rotates within it, so stars appear to rise in the east and set in the west, just like the Sun. An observer on Earth can see a maximum of half of the sphere at one time.

◁ **Plotting the stars**
The Earth's poles and equator are projected onto the celestial sphere. A system of grid lines through the poles, called lines of right ascension, and lines parallel to the equator, called declination lines, means stars can be located by their coordinates.

Spectroscopy

In the laboratory, scientists investigate the chemical elements in hot gases using a technique called spectroscopy. Observing the gases through a spectroscope reveals the different wavelengths of light (see page 196). White light produces a continuous band of colors, but if atoms are present they affect the light and lines of color appear. The atoms of each element have their own unique pattern of lines, called an emission spectrum. Astronomers use spectroscopy to find out what materials are present many light-years away.


emission spectrum of carbon


emission spectrum of hydrogen


emission spectrum of mercury

REAL WORLD

Light-years

Distances in space are so immense that they are measured in light-years, or the distance light travels in a year—slightly more than 9 trillion km (6 trillion miles). The Sun is eight light-minutes away. Voyager 1 (right), the most distant space probe, is 16 light-hours away, while our next nearest star is four light-years out.